

From Specialists to Generalists: Merging LoRA-Adapted Models with DARE and AdaMerging

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1 Introduction

We post-train Qwen3-1.7B across four domains (math, knowledge, multilinguality, safety) to produce four specialists and one generalist. This project aims to solve the catastrophic forgetting inherent in compact models without relying on computationally prohibitive methods. To address this, we propose a modular approach by using LoRA to finetune our models on specific tasks and then merging them with DARE and AdaMerging, while ensuring all models generate verifiable answers enclosed in a `\boxed{}` environment since the output format is strictly constrained by OpenCompass.

2 Methodology

Phase 1: domain dataset construction. Each member builds and curates high-quality domain-specific datasets (math, knowledge, multilinguality, safety) with a standardized output structure, ensuring all answers are enclosed in `\boxed{}` and include structured reasoning in `<think>...</think>`.

Phase 2: domain training (SFT & RL). Each member builds a LoRA adapter for their domain using Low-Rank Adaptation (Hu et al., 2022). The team agrees on rank r , alpha α and identical target modules. This is a critical constraint: task vectors are only additively compatible if the adapter rank, alpha, and target modules are identical across all experts. For the optimization process, each specialist will first undergo Supervised Fine-Tuning (SFT) on domain-specific data. Subsequently, we will apply tailored alignment techniques based on the domain's evaluation nature. For the objective domains (Mathematical Reasoning and General Knowledge), we will utilize Reinforcement Learning with Verifiable Rewards (RLVR), as evaluation relies on exact-match answers. For subjective domains (Multilinguality and Safety), we instead apply Direct Preference Optimization (DPO)

(Rafailov et al., 2023), which directly optimizes preference pairs without requiring a separate reward model or full RL pipeline.

Phase 3: merging. To merge the 4 LoRAs, we will first apply DARE (Yu et al., 2024) to drop negligible weights. We would then use AdaMerging (Yang et al., 2024) to learn per-layer weights. We will keep TIES-Merging on hand as a backup in case of mathematical interference.

Datasets. Each task requires specific datasets: one for the SFT phase and one for the Reinforcement Learning phase, both of which will be explicitly split into training and validation sets. Each dataset is constructed from existing benchmarks and public datasets discussed in literature review, reformatted to our strict specifications. For datasets that lack reasoning traces or boxed outputs, we use a stronger teacher model to generate responses in the expected format.

Evaluation. Model evaluation will be conducted using the OpenCompass framework, utilizing the task-specific benchmarks detailed in the literature review. Finally, we will evaluate the unified model across all four benchmarks, comparing its scores against the individual specialists to quantify catastrophic forgetting and assess our merging strategy

3 Project Timeline

W1 : LoRA config chosen, dataset construction

W2 : SFT (4 domains), initial HF checkpoints

W3 (MS2): RL/DPO finished

W4 : Merging & light calibration

W5 : Generation-config sweep, final report.

Model	Student
Mathematical Reasoning	Julien Erbland
General Knowledge	Max Henrotin
Multilinguality	Mathis Richard
Safety	Morgane Magnin

Table 1: Roles

References

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